Safe System of Work

Isolation and Stored Energy – Vehicles, Plant and Workshop Machinery

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Overview

This Method Statement is intended to provide, as far as is reasonably practicable, a safe system of work for isolating vehicles, trailers, plant and workshop machinery, and making all stored energy safe, before inspection, maintenance, repair, cleaning or fabrication work starts. It details the steps to be taken and the method used to carry out the task safely. All personnel carrying out this activity are expected to co-operate fully with the methods described and follow the risk control measures identified in the associated risk assessments.

Scope

- HGVs, tippers, trailers, sweepers, low loaders, hook-loaders and vehicles fitted with cranes or other hydraulic equipment.
- Mobile plant brought into the workshop or repaired on site.
- Workshop machinery: pillar and radial arm drills, saws, grinders, compressors, vehicle lift tables, gantry crane, washdown equipment and fabrication machinery.
- Work in the workshop, in Fox Group yards, on customer or construction sites, and at the roadside.

Hazards Identified

- Unexpected starting, movement or rolling of a vehicle or machine – crushing, entanglement, run-over
- Gravity – raised tipper bodies, cabs, cranes, attachments or chassis falling
- Hydraulic pressure – sudden movement, hose failure, high-pressure injection injury
- Compressed air – air tanks, brake systems, spring brake chambers and air suspension releasing energy
- Electrical – vehicle 12/24 V systems (burns, arcing, fire), mains-powered machinery (shock), high-voltage electric/hybrid vehicles
- Mechanical – rotating parts that run on (fans, sweeper brushes, flywheels, PTOs), springs under tension
- Thermal – hot engines, exhausts and pressurised cooling systems
- Someone else re-energising equipment while you are working on it

Personal Protective Equipment (PPE) Requirements







Personal Protective Equipment – each fitter / fabricator is equipped with the following:

- Safety Footwear
- Full Length High Visibility Clothing in yards, on site and at the roadside
- Gloves (multi-purpose; chemical-resistant where fluids are present)

- Safety Glasses (face shield when working near pressurised hydraulic or air systems)
- Hard hat where required by site rules or overhead hazards

Equipment or Machinery Used.

- Personal safety padlock with a single key, marked with the fitter's name (one person – one lock – one key)
- Multi-lock hasps, lockable battery isolator covers, plug lock-out boxes, circuit-breaker and valve lock-outs
- "DANGER – VEHICLE UNDER REPAIR – DO NOT START OR MOVE" and "DO NOT OPERATE" tags
- Lockable key safe for vehicle and plant keys
- Wheel chocks, axle stands, body props, cab-tilt supports, crane/boom supports
- Spring brake caging (wind-off) bolts, air-line gauges, approved voltage indicator and proving unit (electrician use)

COSHH Relevant.

Hydraulic oil, brake fluid, coolant and battery acid may be released during isolation. Follow SSoW-070 COSHH – Hazardous Substances in the Workshop and the product COSHH assessments.

Emergency Procedures.

Should an accident or incident occur on Fox Group sites, it will be reported to the site responsible person and the Fox Group accident reporting procedure will be followed. The incident must also be reported through the near miss QR code on the front page of the RAMS and up around Sandham House.

High-pressure hydraulic or air injection injuries, even if they look minor, are a medical emergency – go to A&E immediately and tell them it is an injection injury.

Emergency Contact.

Emergency Contact Details (IN CASE OF EMERGENCY OR INCIDENT)

Fox Group Breakdown – 07923447383

Emergency services – 999 · First aid: see WF041 First Aid · Spills: see WF040 Emergency Spill Response

Description of Activity/Task.

Before anyone works on a vehicle, machine or item of plant, it must be stopped, isolated from all energy sources, and all stored energy must be released or restrained. The isolation must then be secured so it cannot be undone by anyone else, and proved to be effective before work starts.

Emergency stops, interlocks, control switches and simply "turning it off" are NOT isolation. Never rely on hydraulics alone to hold anything up.

The Workshop Supervisor is responsible for issuing personal locks, maintaining lock-out equipment, and authorising isolation of fixed workshop electrical supplies. Work on high-voltage (electric or hybrid) vehicle systems is only to be carried out by persons trained and authorised for that work (see WF021).

Statutory Examinations.

- Vehicle lift tables and vehicle lifts: thorough examination under LOLER by a competent person – six-monthly as recommended by HSE for lifts that people work beneath.
- Gantry crane and lifting accessories (chains, slings, shackles): thorough examination under LOLER – lifting accessories at least every 6 months; other lifting equipment at least every 12 months, or per the examination scheme.
- Trolley and bottle jacks: thorough examination annually; axle stands and body props inspected annually and checked before each use.
- Air receivers and compressors: examined in accordance with the written scheme of examination under the Pressure Systems Safety Regulations 2000.

- Vehicles are subject to 6-weekly inspections and annual tests. Lockable isolators on workshop machinery are checked during the monthly workshop inspection.

METHOD STATEMENT

Key rules

- **One person – one lock – one key.** Every person working on the equipment fits their own personal lock. Never work under someone else's lock.
- **Only the person who fitted a lock removes it.** Keep the key on you at all times.
- **Isolate – lock – tag – release – prove.** Do not start work until every step is complete.
- **If in doubt, stop and ask** the Workshop Supervisor.

Step 1 – Plan the isolation

- Identify every energy source on the vehicle, plant or machine: electrical, hydraulic, pneumatic (air), mechanical, gravity, thermal and fuel. Use the manufacturer's information where available.
- Inform the Workshop Supervisor and anyone else working on or around the vehicle or machine.

Step 2 – Position and secure the vehicle or plant

- Park on level, firm ground in the designated workshop bay or a safe area away from traffic (see WF011 and WF030).
- Apply the parking brake, put the gearbox in neutral and chock the wheels on both sides of an axle that is not being lifted.
- Lower attachments, buckets, forks and cranes to the ground or onto supports.

Step 3 – Switch off and control the keys

- Stop the engine and remove the ignition key. Keep the key on your person or lock it in the key safe using your personal lock.
- Fit the "VEHICLE UNDER REPAIR – DO NOT START OR MOVE" tag to the steering wheel and cab door.
- For keyless or push-button start vehicles, keep the fob well away from the vehicle in the locked key safe.

Step 4 – Isolate electrical energy

- Vehicles: switch off the battery master isolator and lock it with your personal lock where it is lockable. Where there is no lockable isolator, disconnect the battery earth (negative) lead first and insulate it (see WF020).
- Workshop machinery: switch off at the local isolator and lock it in the OFF position with your personal lock and tag. For plug-in equipment, unplug and fit a plug lock-out box, or keep the plug under your control.
- Fixed electrical installations: isolation and proving dead by a competent electrician only.
- Electric/hybrid vehicles: high-voltage isolation only by trained and authorised persons following the manufacturer's procedure (WF021).

Step 5 – Support raised loads (gravity)

- Tipper bodies must be fully lowered, or propped with the correct body prop before anyone goes under them (see WF031 Working Under Propped Body and SSoW-054).
- Tilt cabs must be fully tilted to the locked position with the cab-tilt support engaged.
- Never work under a vehicle supported only on a jack – use axle stands of suitable rated capacity on firm ground (see WF019 and WF026).
- Cranes, booms and hook-loader arms are stowed or supported mechanically. Never rely on hydraulic rams or valves to hold a load.

Step 6 – Release hydraulic pressure

- With the engine stopped, operate the hydraulic controls through their full range to relieve residual pressure, following the manufacturer's procedure. Discharge accumulators where fitted.
- Slacken fittings slowly with a rag over the joint and your face away from the line. Never use your hand to search for leaks – use a piece of card.
- Allow hot hydraulic oil to cool before opening the system.

Step 7 – Release or restrain air pressure

- Before working on brake, suspension or air systems, drain the air tanks and confirm zero pressure on the gauge.
- Spring brake chambers: wind off (cage) the spring using the caging bolt before removing the chamber or releasing the park brake mechanically. Never dismantle a spring brake chamber.
- Air suspension: support the chassis on axle stands before disconnecting air bags or levelling valves.
- Workshop compressors: isolate electrically, close the outlet valve and vent the receiver before working on the compressor or air lines.

Step 8 – Mechanical and thermal energy

- Wait until all rotating parts have stopped – fans, sweeper brushes, PTOs, drill spindles and saw blades can run on.
- Release or restrain springs and tensioned belts in line with the manufacturer's instructions.
- Let engines, exhausts and cooling systems cool down. Do not open a hot pressurised cooling system (WF023).

Step 9 – Prove the isolation

- Try to start the vehicle or machine and operate the controls to prove nothing moves. Return all controls to OFF / neutral afterwards.
- Check gauges read zero pressure and that raised parts are secure on their mechanical supports.
- Only now may work start.

Step 10 – Multiple workers and shift handover

- Where more than one person is working, fit a multi-lock hasp to the isolation point. Each person fits their own lock.
- At shift handover, the incoming fitter fits their lock BEFORE the outgoing fitter removes theirs. Pass on the job status verbally and on the job card.

Step 11 – Return to service

- Check all tools, equipment and people are clear; guards, covers and panels are refitted.
- Remove your own lock and tag. Reconnect batteries and restore supplies in the reverse order.
- Re-pressurise air and hydraulic systems, check for leaks and test the function before the vehicle or machine is released.
- Remove body props, axle stands and chocks only when it is safe to do so. Update the job card and tell the Workshop Supervisor.

Lost key or lock left on

A lock may only be removed by someone other than its owner when ALL of the following are complete:

- The Workshop Supervisor or Workshop Manager has confirmed the lock owner is not on site and has attempted to contact them.
- The supervisor has personally checked that the vehicle or machine is safe and no one is working on it.
- The removal is recorded with the date, time, reason and name of the person authorising it, and the lock owner is told before they next start work.

Mobile, site and roadside repairs

- The same principles apply away from the workshop. Remove and keep the key, fit the tag, chock the wheels and support raised parts before starting work.
- Follow the site's own isolation or permit rules where they exist, and SSoW-047 Use of Vehicles on Construction Sites.
- At the roadside, follow SSoW-012 Changing HGV and Trailer Wheels and Tyres and Fox Group breakdown arrangements.

Under the Health and Safety at Work Act, you are required to inform your Manager of any Health and Safety issues you may have that would affect or endanger yourself or others whilst carrying out your contracted duties.

Related sections

WF004 Electricity · WF006 Lifting Operations · WF010 Use of Vehicle Pit · WF011 Vehicle and Plant Movements · WF019 Working Under Vehicles · WF020 Working with Batteries · WF021 Working on Electric Vehicles · WF022 Repairing Vehicle Brakes · WF023 Working on Cooling Systems · WF026 General Jacking and Blocking · WF027 Checking and Changing Hoses · WF028 Sweeper Vehicles · WF031 Working Under Propped Body · WF033 Vehicle Lift Table · WF034 Radial Arm Drill · WF035 Gantry Crane · WF037 Pillar Drill · SSoW-012 · SSoW-047 · SSoW-054 · SSoW-070 COSHH (draft)

References

- Provision and Use of Work Equipment Regulations 1998 (PUWER), regulation 19 (isolation from sources of energy)
- HSE – Maintenance of work equipment (safe isolation): https://www.hse.gov.uk/work-equipment-machinery/maintenance.htm
- HSE – Vehicle inspection, maintenance and repair: https://www.hse.gov.uk/workplacetransport/vehicles/inspection.htm
- HSE INDG434 – Working safely under motor vehicles being repaired: https://www.hse.gov.uk/pubns/indg434.pdf
- HSE – Working under vehicles (vehicle lift examination): https://www.hse.gov.uk/mvr/mechanical-repair/under-vehicles.htm
- HSE – Thorough examinations and inspections of lifting equipment: https://www.hse.gov.uk/work-equipment-machinery/thorough-examinations-lifting-equipment.htm
- HSE HSG253 – The safe isolation of plant and equipment: https://www.hse.gov.uk/pubns/books/hsg253.htm
- HSE – Electric and hybrid vehicles: https://www.hse.gov.uk/mvr/topics/electric-hybrid.htm